

Networking Projects Report

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Field: Information Technology

Project Type: Cisco Packet Tracer Network Simulation

Introduction

This report presents several networking projects developed using Cisco Packet Tracer. The purpose of these projects is to demonstrate practical knowledge of configuring routers and switches, implementing VLANs, routing protocols, and essential network services. These projects simulate real-world enterprise network environments and strengthen hands-on networking skills.

Tools Used

Cisco Packet Tracer
Cisco IOS CLI
GitHub

Objectives

- Configure Cisco routers and switches.
- Implement VLAN segmentation.
- Configure trunk links.
- Enable inter-VLAN communication.
- Configure DHCP services.
- Implement static routing.
- Configure dynamic routing protocols.
- Verify network connectivity using testing commands.

Project Summaries

Project 1 – VLAN Using Trunk Mode

Objective

Configure VLANs on switches and allow communication between switches through trunk links.

Skills Demonstrated

- VLAN creation
- Access port configuration
- Trunk configuration
- Switch management
- Network segmentation

Devices Used

- Cisco Switches

- PCs

Result

Successfully configured multiple VLANs and enabled trunk communication between switches.

Project 2 – Router-on-a-Stick (ROAS)

Objective

Implement Inter-VLAN Routing using Router-on-a-Stick.

Skills Demonstrated

- Sub-interface configuration
- 802.1Q encapsulation
- Default gateway configuration
- Router configuration

Devices Used

- Router
- Switch
- PCs

Result

All VLANs successfully communicated through a single router interface.

Project 3 – VLAN with DHCP

Objective

Automatically assign IP addresses to hosts within multiple VLANs.

Skills Demonstrated

- DHCP pools
- Gateway configuration
- IP addressing
- VLAN configuration

Result

Hosts automatically received valid IP addresses and communicated successfully.

Project 4 – Static Routing

Objective

Connect multiple networks using manually configured routes.

Skills Demonstrated

- Static Route configuration
- Router interface configuration
- IP addressing
- Network verification

Result

Successful communication between separate networks using static routes.

Project 5 – RIP Routing Protocol

Objective

Configure Routing Information Protocol (RIP) for automatic route sharing.

Skills Demonstrated

- RIP Version 2
- Network advertisement
- Route verification

Result

Routers dynamically exchanged routing information and achieved full connectivity.

Project 6 – OSPF Routing Protocol

Objective

Configure Open Shortest Path First (OSPF) routing.

Skills Demonstrated

- OSPF Area 0
- Router ID configuration
- Neighbor relationships
- Route propagation

Result

Efficient dynamic routing established across all connected routers.

Project 7 – BGP Routing Protocol

Objective

Configure Border Gateway Protocol (BGP) between autonomous systems.

Skills Demonstrated

- BGP neighbor configuration
- Autonomous System (AS) setup
- Route advertisement

Result

Successful exchange of routing information between different autonomous systems.

Skills Gained

Throughout these projects, I gained experience in:

Cisco IOS CLI	Router configuration
Switch configuration	VLAN implementation
Trunk configuration	Inter-VLAN Routing
DHCP configuration	Static Routing
RIP Routing	OSPF Routing
BGP Routing	IP Addressing
Network Troubleshooting	Connectivity Testing
Cisco Packet Tracer Simulation	

Conclusion

These Cisco Packet Tracer projects provided practical experience in designing, configuring, and troubleshooting computer networks. By implementing VLANs, routing protocols, DHCP services, and inter-VLAN communication, I developed a strong foundation in networking concepts and Cisco device configuration. These projects reflect my ability to apply theoretical networking knowledge to simulated enterprise environments and prepare me for roles such as IT Support, Network Technician, Data Center Engineer, and Junior Network Engineer.